

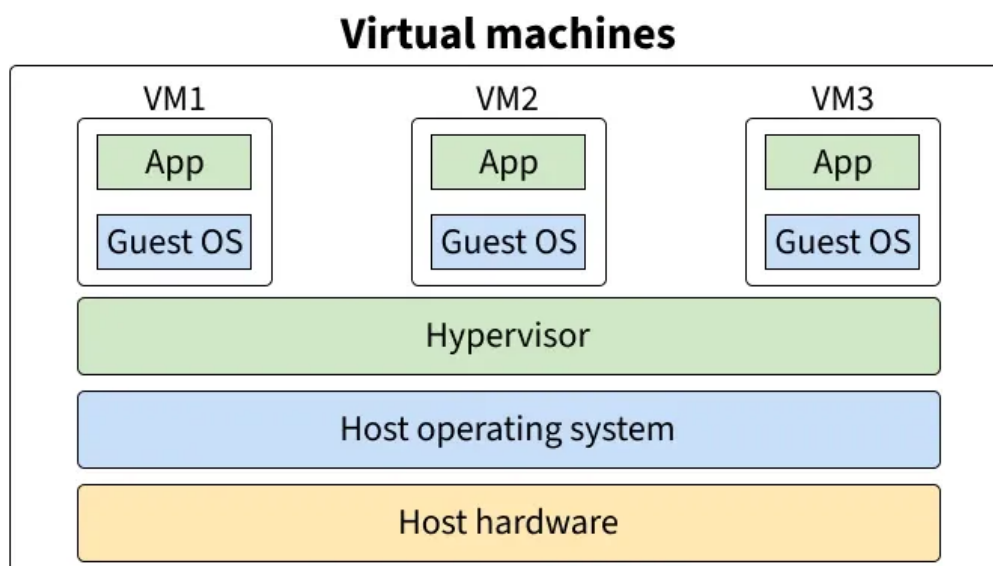
1. Virtualization & Computing Environments

1.1 Virtualization

Virtualization allows multiple operating systems to run concurrently on a single physical machine by abstracting hardware resources.

Types

- **Full Virtualization:** Guest OS runs unmodified (e.g., VMware, VirtualBox).
- **Paravirtualization:** Guest OS is aware of virtualization (e.g., Xen).
- **OS-level Virtualization:** Containers share the host kernel (e.g., Docker).

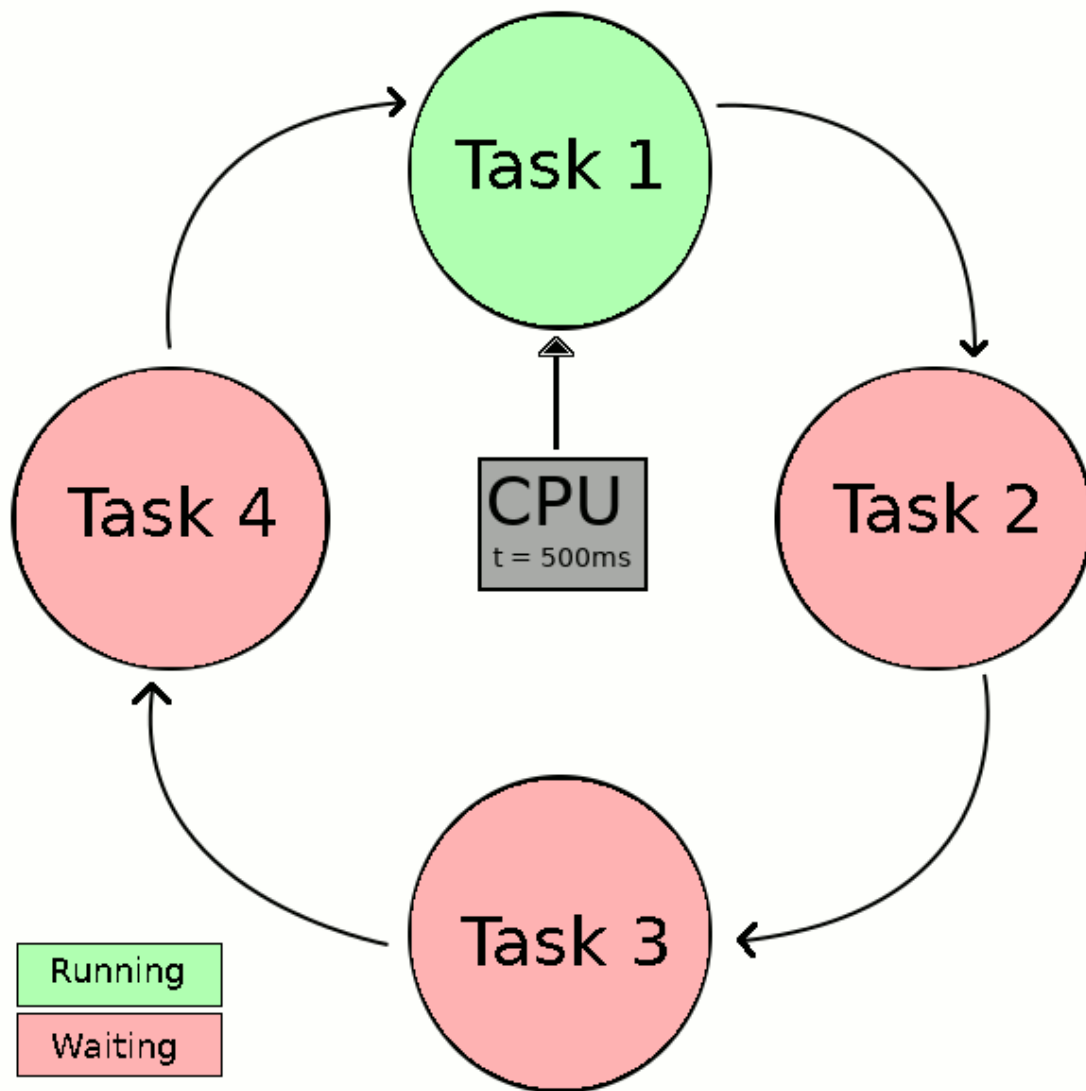


Example: Running Ubuntu and Windows simultaneously on a laptop using VirtualBox.

1.1 Real-Time Embedded Systems

An **embedded system** is a specialized computer designed to perform a **specific task** rather than multiple general-purpose tasks. These systems are commonly found in everyday devices such as **cars, microwave ovens, washing machines, robots, and medical equipment**. They usually run with **limited hardware resources** and have **little or no user interface**, as their main job is to **monitor and control hardware components** like sensors, motors, or engines.

Most embedded systems operate using a **real-time operating system (RTOS)**, where tasks must be completed **within strict time limits**. If the system fails to respond on time, it may lead to serious failures, especially in safety-critical applications like **automobile control systems or industrial robots**. Embedded systems continue to grow in importance as homes, industries, and smart devices become more automated and interconnected.



1.2 Multimedia Systems

Multimedia Systems Most operating systems are designed to handle conventional data such as text files, programs, word-processing documents, and spreadsheets. However, recent trend in technology is the incorporation of multimedia data into computer systems. Multimedia data consist of audio and video files as well as M conventional files. These data differ from conventional data in that multimedia data--such as frames of video--must be delivered (streamed) according to certain time restrictions (for example, 30 frames per second).

Example:

Smartphone

A smartphone is a multimedia system because it handles **audio, video, images, text, and animations**.

Examples:

- Playing videos (YouTube)
- Music playback
- Video calls
- Gaming with sound and graphics

Handheld Systems:

Handheld systems are **small, portable computing devices** such as **smartphones, PDAs, and tablets** that use **embedded or mobile operating systems**. Due to their compact size, these devices have **limited memory, slower processors, and small display screens** compared to desktop computers. Because of this, the operating system and applications must be designed to use **memory and power very efficiently**. Developers must carefully manage resources so that applications run smoothly without wasting memory or battery. Despite these limitations, handheld systems are widely used because they are **lightweight, convenient, and capable of performing essential computing tasks on the go**.

Example:

Tablet (iPad, Android Tablet)

Portable device used for reading, online classes, gaming, and videos.

Computing Environments

So far, we have provided an overview of computer-system organization and major operating-system components. We conclude with a brief overview of how these are used in a variety of computing environments.

Cloud Service Models

1. Infrastructure as a Service (IaaS)

IaaS provides **virtualized hardware resources** like servers, storage, and networks.

Example: AWS EC2

- You rent a **virtual machine** from Amazon.
- You choose the operating system (Linux/Windows), CPU, RAM, and storage.
- You install and manage your own applications.

Use case:

A company launches a website on AWS EC2 instead of buying physical servers.

2. Platform as a Service (PaaS)

PaaS provides a **ready-made platform** to develop and deploy applications.

Example: Google App Engine

- No need to manage OS or servers.
- Developers only upload code.
- The platform handles scaling and maintenance.

Use case:

A developer deploys a Python web app without worrying about server setup.

3. Software as a Service (SaaS)

SaaS provides **fully functional software** over the internet.

Example: Gmail

- No installation required.

- Accessible through a web browser.
- Managed entirely by the service provider.

Use case:

Users send emails using Gmail without managing mail servers.